

Base Clustering Approach and View-Integration Stage in Multiview Entity Resolution: A Controlled Study Across Deduplication and Record Linkage

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Abstract

Multiview entity resolution (ER) must decide whether to integrate several imperfect similarity views before clustering (pre-clustering similarity integration) or to cluster each view and reconcile the partitions afterwards (post-clustering partition integration), and whether the view-specific base clustering is thresholded connectivity or Louvain community detection. We report a controlled study crossing these two design dimensions — base clustering approach (pairwise-threshold vs. Louvain graph) and view-integration stage relative to clustering (pre- vs. post-clustering) — under both ER regimes (deduplication and record linkage), on five benchmark tasks with three views each. Every cell shares a base-configuration grid of 98 candidates; the post-clustering cells additionally select a consensus vote threshold α on development data, a parameter only they possess. Configuration selection uses development-side adjusted Rand index (ARI); the reported outcome is held-out ARI; both are computed by the pinned **GCMER** implementation, and no test label influences any choice. Because the two axes interact, we report conditional contrasts before any pooled effect. The base clustering approach has no material main effect: the evaluated Louvain and threshold-connected-components methods were nearly indistinguishable on four datasets (cell-mean ARI differences ≤ 0.007 , in both directions), their only material separation being on CORA (the benchmark with the fewest truth entities) under pre-clustering integration, where Louvain led by 0.14 ARI — a dataset-dependent interaction, not a stable effect. Pre-clustering integration is robust without any consensus parameter and matched or outperformed calibrated post-clustering integration on four of five datasets (median cell-mean advantage 0.026 ARI over the five datasets); post-clustering integration is competitive when α is calibrated but sensitive to it, collapsing under the fixed majority vote on one coarse-view linkage task and largely recovering once α is selected on development data. The contribution is this controlled factorial evaluation with development-only selection and held-out partition evaluation; its principal limitation is the small dataset count ($n = 5$, the dataset being the independent unit) together with the additional α search, which leaves the pre-versus-post comparison unequally budgeted.

1 Introduction

Entity resolution (ER) identifies the records in one or more sources that refer to the same real-world entity (Fellegi and Sunter, 1969; Christen, 2012a). In practice a record rarely offers a single reliable comparison signal. A bibliographic record exposes a title, an author list, and a venue; a personal record exposes a name, an address, and geographic attributes. Each such group of fields — a *view* — induces its own similarity structure over record pairs, and the views disagree: titles are mistyped, author lists are truncated, venues are abbreviated inconsistently. Multiview ER must therefore answer a design question that arises before any specific matcher is chosen:

Given multiple imperfect views of the same records, should the views be integrated into a single similarity signal *before* clustering, or should each view be clustered separately and the resulting partitions reconciled *afterwards*?

This question — the *view-integration stage relative to clustering*, which we abbreviate to the *integration stage* — interacts with a second one: the *base clustering approach* applied within each view. Pairwise-threshold methods classify record pairs by thresholding similarities and induce clusters through explicit connectivity; graph/community methods treat similarities as a sparse weighted graph and extract clusters with community detection. Both families are common in ER pipelines (Hassanzadeh et al., 2009; Saeedi et al., 2017), yet they respond very differently to noisy edges: a single spurious edge merges two components under transitive closure, whereas community detection can absorb or reject it depending on the surrounding edge density. We evaluate the graph/community family through Louvain modularity optimization, and scope our conclusions about it to that representative rather than to graph clustering in general.

Answers to both questions depend on a third dimension that is fixed by the application: whether the task is *deduplication* (one dirty source, clusters of arbitrary size) or *record linkage* (two sources, matches constrained across sources and dominated by unmatched records). The class imbalance, the truth structure, and the harm of over-merging all differ between the regimes.

A further difficulty is methodological. All four combinations of base clustering approach and integration stage expose tuning parameters — integration weights, thresholds, graph sparsification levels, community resolutions, consensus rules — and published comparisons frequently give different methods different amounts of tuning, or tune on the same labels used for the final report. Model selection with unmatched budgets or leaked labels can reverse observed rankings (Varma and Simon, 2006; Cawley and Talbot, 2010). Whether one integration stage “wins” can therefore be an artifact of the selection protocol rather than a property of the method.

This paper contributes a controlled empirical study of the three-dimensional design space, with matched base-configuration grids and development-only selection. We crossed two base clustering approaches (pairwise-threshold vs. Louvain graph/community) with two integration stages (pre-clustering similarity integration vs. post-clustering partition integration) under both ER regimes, on five benchmark tasks with three views each. Every cell of the design searched the same shared grid of 98 candidate configurations, the same entity-disjoint development/test splits (five per dataset), the same similarity inputs, and the same selection metric — development-side ARI, matching the partition nature of the ER output — computed on development entities only; the consensus threshold, a parameter only the post-clustering pipelines possess, was likewise selected on development data, and we explicitly separate the resulting tuned-method comparison from the fixed-rule sensitivity view. Because the two design axes interact, we report the conditional (within-stage and within-approach) contrasts and their interaction before any pooled main effect. Alongside held-out accuracy we recorded graph-structure diagnostics that make failure mechanisms observable, and outcome-independent view-complementarity descriptors computed on development data only, which we relate to the held-out gain of multiview integration over the best single view.

2 Related Work

Entity resolution and record linkage. Probabilistic record linkage originates with Newcombe et al. (1959) and was formalized by Fellegi and Sunter (1969); Jaro (1989) supplied the string-comparison machinery still in wide use. Surveys cover the field from complementary angles: duplicate detection (Elmagarmid et al., 2007; Naumann and Herschel, 2010), methodology and

practice (Christen, 2012a; Winkler, 2006; Getoor and Machanavajjhala, 2012), and end-to-end pipelines for large-scale settings (Christophides et al., 2020). Benchmark-based comparisons of matching systems were standardized by Köpcke et al. (2010), whose DBLP–ACM task we reuse, and benchmark profiling was refined by Primpeli and Bizer (2020). Modern learned matchers (Mudgal et al., 2018) address pair classification; our study instead concerns the clustering stage that turns pairwise evidence into a partition.

Clustering in ER. Hassanzadeh et al. (2009) evaluated clustering algorithms for duplicate detection over thresholded similarity graphs, and the Rahm group compared clustering schemes for *multi-source* ER within the FAMER framework (Saeedi et al., 2017, 2018). These studies vary the clustering algorithm on a single fused similarity graph; they do not study multiple similarity *views* of the same record pairs, nor the order in which views are clustered and combined. Community-detection algorithms such as Louvain (Blondel et al., 2008) and Leiden (Traag et al., 2019) provide the graph/community family evaluated here.

Multiview learning and clustering. The observation that multiple views of the same objects can be exploited jointly goes back to co-training (Blum and Mitchell, 1998); multiview clustering was introduced by Bickel and Scheffer (2004) and developed through co-regularized spectral methods (Kumar et al., 2011) and a large subsequent literature (Xu et al., 2013; Chao et al., 2021). Within that literature, the contrast between *early* (feature/similarity-level) and *late* (decision/partition-level) integration — with late-fusion multiview clustering combining per-view partitions or base kernels rather than raw features (Liu et al., 2019; Wang et al., 2019; Huang et al., 2023) — is the same dichotomy our second dimension studies. We use the more explicit terms *pre-clustering similarity integration* and *post-clustering partition integration*, which name what is integrated and when relative to clustering. That subfield, however, targets generic vector benchmarks with roughly balanced clusters; ER brings extreme pair imbalance, dominant singleton entities, and transitivity constraints that change how order effects play out. Similarity network fusion (Wang et al., 2014) and consensus clustering (Strehl and Ghosh, 2002; Monti et al., 2003) supply integration and consensus machinery that we adapt: our consensus rule is a weighted co-membership vote in the spirit of cluster ensembles (Strehl and Ghosh, 2002).

Evaluation and model selection. Clustering agreement is measured by ARI (Hubert and Arabie, 1985), information-theoretic indices (Vinh et al., 2010; Meilă, 2007), the Fowlkes–Mallows index (Fowlkes and Mallows, 1983), and extrinsic metrics such as B-cubed (Amigó et al., 2009); Hand and Christen (2018) caution on F-measure use in record linkage, which motivates our reporting of multiple metric families from a single selected configuration. Selection-induced optimism is well documented (Varma and Simon, 2006; Cawley and Talbot, 2010); we control it with entity-disjoint selection/evaluation splits and matched base-configuration grids (with the additional post-clustering α search disclosed explicitly). Blocking and indexing surveys (Christen, 2012b; Papadakis et al., 2020; Draibach and Naumann, 2011) motivate our decision to avoid blocking entirely at the scales studied, so that candidate-generation loss cannot confound the comparison.

Position of this study. Each ingredient of our design exists in isolation — clustering comparisons for ER on fused graphs, feature-level versus partition-level view combination in generic multiview clustering, and consensus ensembles — none of which is itself novel here. What we contribute is a controlled factorial evaluation that crosses the base clustering approach with the integration stage over multiview similarity evidence, under both deduplication and record linkage, with matched base-configuration grids, explicit disclosure of the additional post-clustering α search, development-only configuration selection, and entity-disjoint held-out partition evaluation. This crossed design, rather than any individual axis, is the contribution; we are not aware of a prior study combining these design factors in this way.

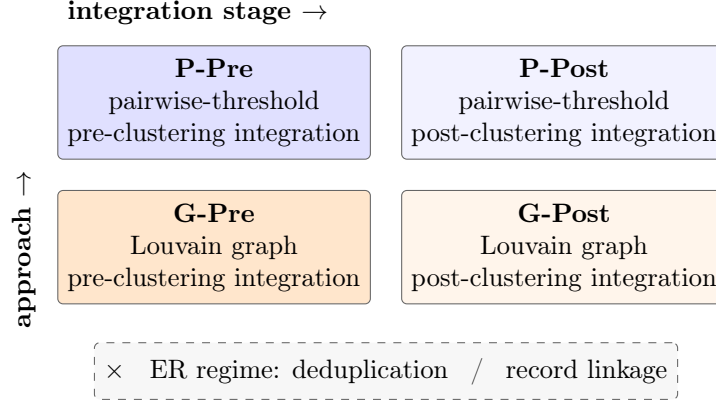


Figure 1: The three-dimensional research framework. Two base clustering approaches are crossed with two view-integration stages; every cell is evaluated under both ER regimes with matched base-configuration grids.

3 Research Framework and Hypotheses

3.1 Three binary dimensions

We study pipelines that resolve n records into a partition using V similarity views. The framework crosses three binary dimensions (Figure 1).

Dimension 1: base clustering approach (P vs. G). *Pairwise-threshold* (P) methods threshold a per-pair similarity and induce clusters by connected components of the resulting match graph. *Graph/community* (G) methods retain a sparse weighted similarity graph and partition it with modularity-based community detection; we instantiate this family with Louvain (Blondel et al., 2008), so all conclusions about “G” are about the evaluated Louvain method rather than graph clustering in general.

Dimension 2: view-integration stage relative to clustering (Pre vs. Post). *Pre-clustering similarity integration* (Pre) combines the per-view similarity matrices into a single fused similarity, then performs one clustering operation. *Post-clustering partition integration* (Post) clusters each view independently and then combines the resulting per-view partitions through a common co-membership consensus procedure. These correspond broadly to the *early* and *late* (decision-level) integration of the multiview literature; we use the pre-/post-clustering terms throughout because they state explicitly what is integrated and when integration occurs relative to clustering.

Dimension 3: ER regime. *Deduplication* resolves one source against itself; *record linkage* matches two sources, with candidate pairs restricted across sources and truth dominated by unmatched records.

The four method cells are denoted **P-Pre**, **P-Post**, **G-Pre**, and **G-Post**: the prefix (P or G) is the view-specific base clustering approach and the suffix (Pre or Post) is the integration stage. In the post-clustering cells the prefix identifies only the base clustering algorithm applied within each view; the final partition-consensus procedure — the weighted co-membership vote, its consensus definition, and its α -selection policy — is identical in P-Post and G-Post.

Table 1: Datasets, truth structure, and view definitions.

Dataset	Regime	Records	Entities	True pairs	Views
CORA	dedup.	1,815	118	64,578	title, authors, venue
FEBRL1	dedup.	1,000	500	500	name, address, geo
DBLP-ACM	linkage	4,910	2,690	2,220	title, authors, venue
Restaurant	linkage	864	754	110	name, address, phone
NCVR-2P	linkage	4,000	3,696	304	name, suburb, postcode

3.2 Hypotheses

The hypotheses were fixed before the test results were computed; the study is designed so that each can be rejected.

Throughout, ER partition accuracy is operationalized primarily by the adjusted Rand index (ARI) between the predicted and the true partition of held-out entities; pairwise F_1 is reported as a supplementary operational metric only.

- H1** Under matched base-configuration grids, the base clustering approach affects held-out ER accuracy: pairwise-threshold and Louvain graph/community clustering differ systematically in held-out ARI, possibly conditionally on the integration stage.
- H2** The integration stage interacts with view complementarity: post-clustering pipelines lose held-out ARI when views are individually weak but complementary, because per-view partitions commit to errors that consensus cannot undo.
- H3** The effects of the base clustering approach and the integration stage differ between deduplication and record linkage.
- H4** Where the Louvain graph approach degrades, the degradation is explained by false-edge connectivity: retained spurious edges connect records of different entities into large, impure components.
- H5** Outcome-independent complementarity descriptors computed on development data are associated with the held-out ARI gain of multiview integration over the best single view.

4 Data

We use five benchmark tasks: two deduplication tasks (CORA, FEBRL1) and three record-linkage tasks (DBLP-ACM, Restaurant, NCVR-2P). Four are long-established public benchmarks; NCVR-2P is a fixed, prepared NCVR-derived two-party linkage task (described below). Table 1 summarizes the record universes, truth structures, and view definitions; all statistics were derived directly from the data and truth files, whose SHA-256 checksums are pinned in the released protocol.

CORA (McCallum et al., 2000) contains 1,879 bibliographic records. The truth file encodes 64,578 matched pairs over 1,815 records; the truth graph is transitively closed (every entity forms a complete clique of pairs) and comprises 118 entities (sizes 2–238). The 64 records absent from the truth graph cannot be confirmed as singletons, so the evaluation universe is the 1,815 labeled records. Views: title, authors, and venue, where venue coalesces the journal field (if non-blank) with the book-title field.

FEBRL1 (Christen, 2008) is a synthetic person-record benchmark of 1,000 records: 500 originals, each with exactly one corrupted duplicate. Views: name (given and surname), address (street number and two address lines), and geography (suburb, postcode, state). Social-security number and date of birth are excluded from all comparisons as near-unique identifiers (550 and 514 distinct values per 1,000 records, respectively).

DBLP-ACM (Köpcke et al., 2010) links 2,616 DBLP records to 2,294 ACM records; the truth is one-to-one with 2,220 matched pairs and 470 unmatched records. Views: title, authors, and venue-year. Comparison is full bipartite (6,001,104 candidate pairs); no blocking is applied.

Restaurant (Fodors-Zagat; Tejada et al., 2001) links 533 Fodors records to 331 Zagat records with 110 one-to-one matches and 644 unmatched records. Views: name, address (street and city), and phone (digits only). Comparison is full bipartite (176,423 pairs). Source strings carry markup artifacts that are removed by normalization.

NCVR-2P is a fixed, prepared two-party linkage task derived from the North Carolina Voter Registration data with synthetically corrupted attribute values, in the tradition of NCVR-based linkage benchmarks (Christen and Vatsalan, 2013; Christen, 2014; Panse et al., 2021): 2,000 records per party, 304 entities present in both parties (one record per entity per party), and 3,392 party-exclusive records. Unlike the four public benchmarks above, this exact 4,000-record snapshot ships with the study as a prepared, checksum-pinned file. Its exact subset-construction procedure — which two parties were used, how records and matched entities were sampled, and any random seed — is not recorded and could not be reconstructed, and the file is not a clean subset of any bundled multi-party NCVR source; we therefore treat it as a fixed prepared task rather than an independently reproducible public benchmark, and results on it are reproducible only from the released file (Section 11). Views: name (given and surname), suburb, and postcode. Comparison is full bipartite (4,000,000 pairs); at this scale exhaustive comparison remains feasible, so pair completeness is 1.0 by construction for all five tasks.

Missing view values are never imputed. Truth-bearing columns are excluded from all similarity computations, and for linkage tasks the source attribute is used only to restrict candidate pairs across sources.

5 Methods

5.1 Views and similarities

Text is normalized by lower-casing, removing markup characters, mapping non-alphanumeric characters to spaces, and squeezing whitespace; phone and postcode fields retain digits only. Empty values become missing. Per-view per-pair similarities in $[0, 1]$ use three fixed functions chosen a priori by field type: cosine similarity over character 3-grams for long text views, Jaro-Winkler (prefix scaling $p = 0.1$) for short name-like views, and normalized Levenshtein for digit strings; these function constants are implementation defaults, fixed identically for all cells and never varied. No corpus-level statistics (such as inverse document frequencies) are used, so per-pair similarities are independent of any data split by construction. If either record of a pair lacks a view, that view’s similarity for the pair is missing.

5.2 Integration and consensus

A weight vector w over views drives both integration stages. For pre-clustering integration, the fused similarity of a pair is the weighted mean of the view similarities observed for both records, with weights renormalized over the observed views; pairs with no observed view receive no edge. For

Table 2: Method cells and matched base-configuration grids (the post-clustering cells additionally search four consensus thresholds α). Weight vectors span the uniform vector and six a priori asymmetric vectors; thresholds $t \in \{0.30, 0.35, \dots, 0.95\}$; sparsifications $s \in \{0.35, 0.45, \dots, 0.95\}$; resolutions $\gamma \in \{1.0, 2.0\}$.

Cell	Base clustering approach	Integration stage	Search space (98 configurations)
P-Pre	pairwise-threshold (P)	pre-clustering (similarity)	7 weights $\times$ 14 thresholds
P-Post	pairwise-threshold (P)	post-clustering (partition)	7 weights $\times$ 14 thresholds
G-Pre	Louvain graph (G)	pre-clustering (similarity)	7 weights $\times$ 7 sparsif. $\times$ 2 resol.
G-Post	Louvain graph (G)	post-clustering (partition)	7 weights $\times$ 7 sparsif. $\times$ 2 resol.

post-clustering integration, each view is clustered independently (records lacking the view remain singletons in that view’s partition) and a pair’s consensus vote is the weighted mean of per-view co-membership indicators over views observed for both records, renormalized identically; pairs with vote $\geq \alpha$ form edges and the final partition is the connected components of the vote graph. The consensus threshold α is a genuine parameter of the post-clustering pipelines — pre-clustering pipelines have no analogue — and is treated as one: $\alpha \in \{0.50, 0.60, 0.67, 0.75\}$ is selected on development data jointly with the configuration (Section 6), and the fixed majority-vote rule $\alpha = 0.5$ is analyzed separately as a sensitivity and mechanism case (Sections 8–9). The consensus procedure is identical for the two post-clustering cells, so P-Post and G-Post differ only in their view-specific base clustering algorithm. Using the same weight vectors and the same missing-data rule in both stages keeps the two dimensions comparable.

5.3 Method cells

P-Pre thresholds the fused similarity at t and returns connected components. **P-Post** thresholds each view’s similarity at t , clusters each view by connected components, and applies weighted co-membership consensus. **G-Pre** retains candidate edges with fused similarity $\geq s$ (edge weight equal to the fused similarity) and partitions the graph with Louvain community detection (Blondel et al., 2008) at resolution γ . **G-Post** builds a per-view graph with the same sparsification s , runs Louvain per view at resolution γ , and applies the same weighted co-membership consensus. All community detection uses seeded random states and is exactly reproducible.

5.4 Baseline

The single-view baseline (SV) thresholds one view’s similarity and takes connected components; the best view–threshold combination is selected on development entities exactly as for the multiview cells.

6 Experimental Protocol

The protocol was frozen before the full experiment was executed. A *configuration* is one fully specified pipeline: a method cell, a weight vector, and the cell’s clustering parameters. Each cell shares the same base-configuration grid of exactly 98 candidates per dataset (Table 2); the post-clustering cells additionally search four consensus thresholds α , so their effective selection space is 98×4 candidates (detailed below). The threshold and sparsification grids extend to 0.95 so that selections on tasks with concentrated similarity distributions are not artificially capped by the grid ceiling.

Table 3: Classification of all protocol quantities. Tuned hyperparameters are selected independently per (dataset, split, cell) on that split’s development side only, and the selected values are applied once to the test side without reselection.

Class	Quantities	Rule
Design constants (frozen a priori)	view definitions; text normalization; per-view similarity functions; exhaustive candidate-pair policy (no blocking); entity-disjoint split construction, seeds, and 60/40 ratio; the candidate grids themselves; selection metric (development ARI) and deterministic tie-break chain; consensus vote form; pinned metric implementation; Louvain seeding scheme	fixed by the frozen protocol; never selected on data
Tuned hyperparameters	weight vector w (7 candidates, all cells); threshold t (14, P cells); sparsification s (7, G cells); resolution γ (2, G cells); consensus threshold α (4, post-clustering cells only, tuned-method estimand); single-view baseline (view, t)	selected by development ARI per (dataset, split, cell); reused once on test
Nuisance constants	Jaro–Winkler prefix scaling $p = 0.1$; character 3-gram length; Louvain implementation internals	implementation defaults, identical in every cell; never varied
Sensitivity-only settings	fixed majority-vote rule $\alpha = 0.5$ (fixed-rule estimand); the α -sweep surface; the percolation path (equal weights, $\gamma = 1.0$)	analyzed separately; never drive primary (tuned-method) selection

Fixed versus tuned quantities. Table 3 classifies every quantity of the protocol into one of four roles, so that no parameter is described loosely as “fixed” or “tuned”: design constants frozen in the protocol before execution; hyperparameters selected on development data; nuisance implementation constants never varied; and sensitivity-only settings that never drive primary selection.

Search-space comparability. The shared base-configuration grid equalizes the candidate counts for weight vectors and clustering hyperparameters across the four cells; it does not establish complete fairness, because the post-clustering pipelines possess an additional parameter — the consensus threshold α — that the pre-clustering pipelines do not have. Allowing α to be selected on development data (four values) enlarges the effective selection space of P-Post and G-Post to 98×4 candidates. Consequently the within-stage approach contrasts (P-Pre vs. G-Pre, 98 candidates each; P-Post vs. G-Post, 98×4 each) have matched total candidate counts, whereas the pre-versus-post (integration-stage) contrasts do not, because each post-clustering cell searches four times as many candidates as its pre-clustering counterpart. We therefore report two deliberately distinct estimands and never mix them: the *tuned-method* comparison, in which P-Post and G-Post use the development-selected α — the primary scientific comparison of the integration stage, since α is a material parameter of those pipelines — and the *fixed-rule* comparison at $\alpha = 0.5$, retained as a sensitivity and mechanism analysis of the majority-vote rule. The two answer different questions (“a calibrated post-clustering pipeline” versus “the default majority-vote post-clustering pipeline”), and all results must be read with this search-space asymmetry in mind; we do not artificially replicate pre-clustering configurations merely to equalize counts. Throughout, the matched budget is a matched *count of candidate configurations* evaluated per cell on identical similarity inputs,

splits, and selection metric; it is not a CPU-time or memory match — per-configuration cost differs across cells (Table 10) — nor, as just described, a match of the post-clustering cells’ effective selection space.

Entity-disjoint splits. For each dataset we drew five deterministic entity-disjoint splits (seeds fixed in the protocol). Entities — truth clusters, including unmatched singletons in the linkage regime — are assigned greedily, largest first, to a development side (60% of records) or a test side (40%), so no entity contributes records to both sides. All records of a linked cross-source entity stay on one side. Clustering is executed separately on each side’s records, and candidate pairs never cross sides.

Selection. For every dataset, split, and cell, all configurations are evaluated on the development side, and one is selected by development ARI with a deterministic tie-breaking chain (higher NMI, then lower VI, then higher FMI, then lower over-merge rate, then smaller threshold or sparsification, then smaller resolution, then earlier weight vector, then — for the post-clustering cells — smaller α). For P-Pre and G-Pre the candidate set is the 98-configuration grid; for P-Post and G-Post it is the grid crossed with $\alpha \in \{0.50, 0.60, 0.67, 0.75\}$, so the configuration and the consensus threshold are selected jointly, on development data only. The selected configuration (with its selected α , where applicable) is then evaluated once on the test side. No test-side record, pair, or label influences any weight, threshold, graph parameter, consensus parameter, or method choice, and no test-side metric of any kind is used for selection — in particular, α is never chosen from test results. All reported test metrics for a row come from that single selected configuration. The single-view baseline is selected by the same rule.

Metrics. Because the system output is an entity partition rather than a set of independent binary pair decisions, the primary metric — for configuration selection on development entities and for the reported held-out outcome — is the adjusted Rand index (Hubert and Arabie, 1985) between the predicted and the true partition. ARI evaluates agreement between complete partitions and is adjusted for chance, whereas pairwise F_1 scores record pairs and thereby weights larger clusters through their quadratic pair counts; because the system output is a partition, ARI is the appropriate primary outcome and F_1 is supplementary. As secondary partition metrics we report NMI in the normalization $2I/(H_x + H_y)$ and the variation of information (VI) in bits (Vinh et al., 2010; Meilă, 2007); the main performance table (Table 4) leads with ARI and these two secondary indices. As supplementary pair-level diagnostics we report the Fowlkes–Mallows index and pairwise precision, recall, and F_1 over all record pairs implied by the partitions of the evaluated side (computed exactly from contingency tables), all collected in the supplementary Table 5; these diagnostics never drive selection or primary conclusions and are used only to interpret pair recovery. All partition metrics are computed by the pinned `GCIMER` implementation ($\text{ARI} = \text{GCIMER}::\text{adj_rand}$); the pipeline has no alternative ARI implementation. The adjusted Rand index is undefined (the indeterminate form $0/0$) only when *both* partitions are trivial in the same way — both a single cluster, or both all singletons — as verified directly against `GCIMER::adj_rand`; a non-trivial reference partition therefore yields a defined ARI for *every* predicted partition, including all-singleton or single-cluster predictions. On every development and test side of every split, the restricted truth partition is non-trivial — it contains more than one entity and at least one multi-record entity, so it is neither a single cluster nor all singletons — and the ARI was therefore defined for every evaluation in the study. For the primary experiment this covers all 9,925 evaluated partitions (9,800 development grid points — 98 configurations $\times$ 4 cells $\times$ 25 dataset-split pairs — plus 125 held-out evaluations: 100 selected cell configurations and 25 selected single-view baselines); the single-view development sweep (1,050 evaluations) and the consensus-threshold sweep of Section 9 likewise produced no undefined value. The only degenerate predictions were 15 of the 9,800 development grid points

whose predictions were all singletons ($\text{ARI} = 0$, not undefined). We additionally report predicted-cluster counts, the largest predicted cluster, over-merge and over-split rates, and runtime. For graph cells we record per-configuration diagnostics: retained edges, graph density, connected components, largest-component ratio, component purity, and the precision and recall of the retained edge set. Graph density is the number of retained edges divided by the candidate-pair count of the evaluated side — $m(m - 1)/2$ for deduplication with m side records, and $n_A \times n_B$ for record linkage with per-source side counts n_A, n_B .

Statistical reporting. The dataset/task is the independent unit ($n = 5$). The five repeated entity-disjoint splits of a dataset are dependent, overlapping resamples of one finite record universe; they quantify within-task variability and are summarized as mean, standard deviation, and range — not as independent-sample confidence intervals. Cross-dataset summaries report the mean, the median, and a leave-NCVR-out sensitivity (NCVR-2P produces the largest effects in this study and is not allowed to dominate headline averages).

Reproducibility. Similarities are computed once per dataset and cached; every stochastic step is seeded; results are written atomically with per-dataset checkpoints. An automated validation suite checks base-configuration-grid matching, split disjointness, metric bounds, selection integrity, and exact reproducibility of a random sample of configurations before any interpretation.

7 Results

All 24,500 distinct development evaluations — per dataset and split: 98 configurations for each pre-clustering cell and 98×4 configuration-threshold candidates for each post-clustering cell, of which the $\alpha = 0.5$ slice constitutes the 9,800-point primary development grid of Section 6 — and all held-out evaluations completed without a single failed configuration. Automated validation confirmed the shared base-configuration grids, entity-disjoint splits, selection integrity, metric bounds, the graph-density definition, and exact reproducibility of a random sample of configurations before any number below was interpreted. Unless a passage explicitly says otherwise, P-Post and G-Post results below use the development-selected consensus threshold (the tuned-method estimand); fixed-rule ($\alpha = 0.5$) results are always labeled as such.

7.1 Held-out performance by method cell

Table 4 and Figure 2 report held-out ARI of the development-selected configuration of each cell, averaged over the five splits. Three patterns stand out. First, the pre-clustering cells are strong everywhere: P-Pre reaches ARI 0.760–0.994 and G-Pre 0.899–0.995. Second, G-Pre matches or exceeds P-Pre on every dataset — the two cells select operating points that yield identical partitions on DBLP-ACM and NCVR-2P, differ by at most 0.002 on FEBRL1 and Restaurant, and separate only on CORA (0.899 vs. 0.760). Third, with the consensus threshold selected on development data, the post-clustering cells are competitive on every dataset (P-Post 0.733–0.969, G-Post 0.732–0.976): their largest deficit is on NCVR-2P (0.837 and 0.851 vs. 0.952), where development selection chose $\alpha = 0.60$ in every split and cell. The development-selected α is task-dependent — 0.50 on CORA, FEBRL1, and Restaurant in most splits, 0.67 on DBLP-ACM, 0.60 on NCVR-2P — and under the fixed majority-vote rule ($\alpha = 0.5$) the same cells fall to 0.124 and 0.136 on NCVR-2P, a sensitivity analyzed mechanistically in Section 8 and quantified in Section 9.

Table 4: Held-out partition performance by dataset and method cell, led by the primary metric (ARI) and the secondary indices (NMI, VI). Mean over five entity-disjoint splits; ARI shown $\pm$ split standard deviation (within-task dispersion, not an inferential interval); all metrics of a row come from the single development-selected configuration. For P-Post and G-Post the consensus threshold is selected on development data; the α column gives the modal selected value (α does not apply to P-Pre/G-Pre). Bold marks the best cell per dataset. Supplementary pair-level diagnostics appear in Table 5.

Dataset	Cell	α	ARI	NMI	VI
CORA	P-Pre	–	0.760 ± 0.041	0.930	0.593
	P-Post	0.50	0.733 ± 0.047	0.919	0.691
	G-Pre	–	0.899 ± 0.048	0.962	0.335
	G-Post	0.50	0.732 ± 0.048	0.920	0.680
FEBRL1	P-Pre	–	0.994 ± 0.005	0.999	0.010
	P-Post	0.50	0.969 ± 0.013	0.996	0.054
	G-Pre	–	0.995 ± 0.006	1.000	0.006
	G-Post	0.50	0.976 ± 0.008	0.998	0.036
DBLP-ACM	P-Pre	–	0.916 ± 0.048	0.996	0.082
	P-Post	0.67	0.923 ± 0.015	0.995	0.103
	G-Pre	–	0.916 ± 0.048	0.996	0.082
	G-Post	0.67	0.919 ± 0.025	0.995	0.100
Restaurant	P-Pre	–	0.991 ± 0.012	1.000	0.003
	P-Post	0.50	0.962 ± 0.029	0.999	0.018
	G-Pre	–	0.993 ± 0.010	1.000	0.003
	G-Post	0.50	0.970 ± 0.014	0.999	0.014
NCVR-2P	P-Pre	–	0.952 ± 0.005	0.999	0.013
	P-Post	0.60	0.837 ± 0.016	0.998	0.043
	G-Pre	–	0.952 ± 0.005	0.999	0.013
	G-Post	0.60	0.851 ± 0.031	0.998	0.040

7.2 Paired comparisons: the approach effect depends on the integration stage, and the stage effect interacts with consensus calibration

Because all cells share splits, comparisons are paired within dataset and split. Since the two design axes are not independent, we report the 2×2 decomposition (Table 6) — the four cell means, the *conditional* approach and integration-stage contrasts, and their interaction — before any pooled main effect. Table 7 then gives the order and approach summaries under both estimands, and Figure 3 the split-level dispersion. The within-stage contrasts (P-Pre vs. G-Pre; P-Post vs. G-Post) and the within-approach contrasts (P-Pre vs. P-Post; G-Pre vs. G-Post) are controlled comparisons of the implemented methods: the cells use the same datasets, views, splits, selection metric, and matched configuration counts, but equal configuration counts do not imply identical model flexibility, because threshold-connected components and Louvain use different clustering mechanisms and parameter semantics. The approach contrasts therefore compare the implemented threshold-connected-components method and the evaluated Louvain-based graph method — not pairwise and graph clustering in general — and every conclusion below is scoped to these two implementations. The diagonal comparisons P-Pre vs. G-Post and P-Post vs. G-Pre change both factors and are never read as a main effect.

Approach $\times$ stage interaction (RQ1). The base clustering approach behaves differently under the two integration stages, so its effect must be stated conditionally. Within pre-clustering

Table 5: Supplementary pair-level diagnostics (Fowlkes–Mallows index and pairwise precision, recall, F_1) for the same development-selected configurations as Table 4. These are supplementary only: they never drive selection or primary conclusions, and pairwise F_1 over-weights larger clusters through their pair counts.

Dataset	Cell	FMI	Precision	Recall	F_1
CORA	P-Pre	0.800	0.648	0.990	0.782
	P-Post	0.779	0.618	0.985	0.758
	G-Pre	0.911	0.840	0.991	0.908
	G-Post	0.779	0.615	0.988	0.757
FEBRL1	P-Pre	0.994	0.996	0.992	0.994
	P-Post	0.969	0.986	0.953	0.969
	G-Pre	0.995	0.992	0.998	0.995
	G-Post	0.977	0.978	0.975	0.976
DBLP-ACM	P-Pre	0.918	0.886	0.953	0.916
	P-Post	0.924	0.932	0.916	0.923
	G-Pre	0.918	0.886	0.953	0.916
	G-Post	0.920	0.917	0.925	0.920
Restaurant	P-Pre	0.991	0.983	1.000	0.991
	P-Post	0.963	0.974	0.955	0.962
	G-Pre	0.993	0.991	0.995	0.993
	G-Post	0.971	0.974	0.968	0.970
NCVR-2P	P-Pre	0.953	0.927	0.979	0.952
	P-Post	0.846	0.979	0.731	0.837
	G-Pre	0.953	0.927	0.979	0.952
	G-Post	0.853	0.810	0.900	0.851

integration, the paired P-Pre–G-Pre contrast has dataset-level mean -0.028 and median -0.001 ARI with every per-dataset difference ≤ 0 (identical partitions on DBLP–ACM and NCVR-2P; -0.001 FEBRL1, -0.002 Restaurant, -0.139 CORA). Within post-clustering integration the paired P-Post–G-Post contrast is negligible everywhere (mean -0.005 , median -0.008 , range $[-0.015, 0.004]$). The approach $\times$ stage interaction $(\text{P-Pre} - \text{G-Pre}) - (\text{P-Post} - \text{G-Post})$ therefore has cross-dataset mean -0.023 and median $+0.006$, but it is dataset-dependent rather than a stable cross-dataset interaction: it is carried almost entirely by CORA (-0.140), with the remaining four datasets close to zero ($[-0.004, +0.015]$); the equivalent form $(\text{P-Pre} - \text{P-Post}) - (\text{G-Pre} - \text{G-Post})$ agrees to machine precision. The reading is that the approach effect is not universal: the two implemented methods are interchangeable except when integration happens before clustering on CORA, the benchmark with the fewest truth entities, where community structure survives the selection split better (Section 9). We therefore do not report a single P-versus-G conclusion, and we treat this single-dataset interaction as illustrative rather than as a general effect.

Integration stage (RQ2), tuned-method estimand. With the consensus threshold selected on development data, pre-clustering matched or outperformed post-clustering integration on four of five datasets. The per-dataset mean P-Pre–P-Post differences are $+0.027$ (CORA), -0.008 (DBLP–ACM), $+0.025$ (FEBRL1), $+0.115$ (NCVR-2P), and $+0.029$ (Restaurant); G-Pre–G-Post: $+0.167$, -0.004 , $+0.019$, $+0.101$, and $+0.023$. The cross-dataset medians are $+0.027$ and $+0.023$ (sign consistency 0.8; means excluding NCVR-2P 0.018 and 0.051): a calibrated post-clustering pipeline is typically a couple of hundredths of ARI behind, can essentially tie or marginally lead on a task whose views are individually strong (DBLP–ACM), and retains a material deficit of about

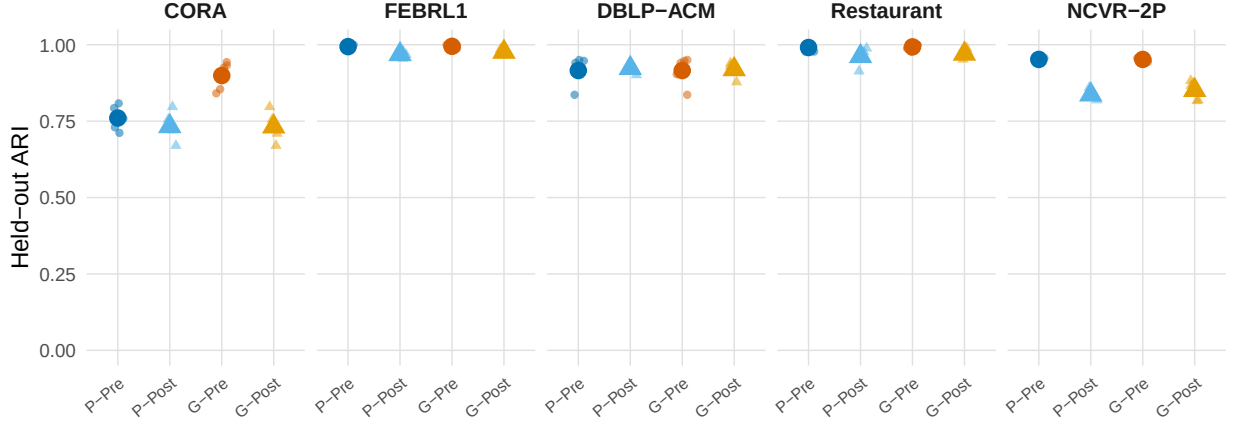


Figure 2: Held-out ARI by method cell and dataset (P-Post/G-Post with development-selected consensus threshold). Small points are the five splits (within-task dispersion); large points are split means. Blue = pairwise-threshold, orange = Louvain graph; circles = pre-clustering, triangles = post-clustering.

0.10 on the coarse-view task even at its best development-selected threshold.

Integration stage, fixed-rule sensitivity. Under the fixed majority-vote rule ($\alpha = 0.5$) the same contrasts are unanimous (sign consistency 1.0) and far larger in the mean (P-Pre–P-Post 0.180, G-Pre–G-Post 0.205) because NCVR-2P contributes 0.828 and 0.817; the medians (0.025, 0.018) are nearly unchanged. The gap between the two estimands is thus almost entirely one dataset’s consensus-rule failure (Section 8); treating the fixed rule as the method would overstate the general stage effect.

7.3 Regime differences (RQ3)

Under the tuned-method estimand the cell-mean integration-stage effect is positive in both regimes — deduplication 0.060 (CORA 0.097, FEBRL1 0.022) and record linkage 0.043 (DBLP-ACM –0.006, NCVR-2P 0.108, Restaurant 0.026) — and the linkage mean is dominated by NCVR-2P: excluding it, the remaining two linkage tasks average 0.010, below the deduplication values. The cell-mean approach effect is small in both regimes (–0.037 deduplication, –0.003 linkage). With two deduplication and three linkage datasets we report these as descriptive differences and do not claim a stable approach×stage×regime interaction. What the design does establish is that the conditions producing a large integration-stage effect (structurally coarse views, most damaging under a majority-vote consensus) arose here in a linkage task dominated by unmatched records, and the conditions producing a visible approach difference (threshold overfitting on few, large entities) arose in a deduplication task.

7.4 Multiview integration versus the best single view (RQ4)

Table 9 compares the development-selected best multiview cell against the development-selected best single view on the same test entities. The held-out ARI gain spans three orders of behavior: +0.383 on NCVR-2P (no single view approaches usable accuracy alone: best single-view ARI 0.57), +0.089 and +0.028 on FEBRL1 and Restaurant (moderate, consistently positive across splits), –0.004 on DBLP-ACM (title alone is essentially sufficient), and –0.054 on CORA. Across datasets the mean gain is 0.088 and the median 0.028; excluding NCVR-2P, mean 0.015 and median 0.012. The CORA loss is dominated by selection noise rather than an intrinsic multiview failure:

Table 6: Factorial decomposition under the tuned-method estimand. (a) Held-out ARI cell means per dataset. (b) Cross-dataset conditional contrasts and the approach $\times$ stage interaction ($n = 5$ datasets; dataset is the independent unit): all-five-dataset mean, median, range of dataset means, and sign consistency (fraction of datasets whose contrast shares the sign of the cross-dataset mean; exact-tie datasets count against consistency). The approach effect appears only within pre-clustering integration and only on CORA. For comparison, the corresponding leave-NCVR-out means are 0.018 (P-Pre–P-Post), 0.051 (G-Pre–G-Post), -0.036 (P-Pre–G-Pre), -0.003 (P-Post–G-Post), and -0.033 (interaction), so the conditional contrasts and the interaction are not driven by NCVR-2P.

<i>(a) Held-out ARI cell means (tuned-method estimand)</i>				
Dataset	P-Pre	P-Post	G-Pre	G-Post
CORA	0.760	0.733	0.899	0.732
FEBRL1	0.994	0.969	0.995	0.976
DBLP-ACM	0.916	0.923	0.916	0.919
Restaurant	0.991	0.962	0.993	0.970
NCVR-2P	0.952	0.837	0.952	0.851
<i>(b) Cross-dataset contrasts ($n = 5$): mean, median, range, sign cons.</i>				
Contrast	Mean	Median	Range	Sign
P-Pre – P-Post (stage P)	0.038	0.027	[-0.008, 0.115]	0.80
G-Pre – G-Post (stage G)	0.061	0.023	[-0.004, 0.167]	0.80
P-Pre – G-Pre (approach pre)	-0.028	-0.001	[-0.139, 0.000]	0.60
P-Post – G-Post (approach post)	-0.005	-0.008	[-0.015, 0.004]	0.60
Approach $\times$ stage interaction	-0.023	0.006	[-0.140, 0.015]	0.40

development ARI picked G-Pre in four of five splits, where the test gain is $+0.001$ on average, but the one split that picked P-Post lost 0.271, and the strong title view leaves integration little to add.

7.5 Computational cost and failures

Mean per-configuration runtime ranged from 0.002 s (Restaurant, P-Pre) to 0.82 s (NCVR-2P, G-Pre); the full study (24,500 development evaluations and all associated held-out evaluations, over up to 6.0×10^6 candidate pairs per dataset) required under two hours on one workstation (Table 10). No configuration failed or exceeded the runtime limit.

8 Mechanism and Complementarity Analysis

8.1 Percolation on the sparsification path — and fragmented selected graphs (RQ5, H4)

Figure 4 traces the equal-weight fused graph on each development side as the sparsification threshold rises, and Table 11 reports diagnostics of the selected graph configurations on test. Two distinct facts must be separated. First, *percolation along the lower-threshold path is real*: on NCVR-2P the fused graph is fully percolated for $s \leq 0.55$ — the largest connected component contains essentially all records (LCC ratio 1.0) with component purity ≈ 0.001 — because Jaro–Winkler similarities between unrelated names, suburbs, and postcodes concentrate well above 0.5. The graph then fragments progressively: the largest-component ratio falls from 0.97 at $s = 0.65$ to 0.115 at $s = 0.75$ and 0.0015 at $s = 0.85$, with high component purity reached only between $s = 0.75$ and $s = 0.85$. Second, *the selected graphs do not operate in the percolated region*. On every dataset,

Table 7: Dataset-level paired held-out ARI differences (each dataset contributes its mean over five splits; $n = 5$ datasets), reported separately for the two estimands. “Cell mean” quantities average the two cells of a dimension before differencing and are labeled explicitly to avoid confusion with the paired contrasts. Reported are the cross-dataset mean, median, leave-NCVR-out mean, and range of dataset means; sign consistency is the fraction of datasets sharing the majority sign. Split-level dispersion appears in the per-dataset Table 8.

Quantity	Mean	Median	Mean excl. NCVR	Range	Sign cons.
<i>Primary: P-Post/G-Post with development-selected consensus threshold</i>					
P-Pre – P-Post (stage P)	0.038	0.027	0.018	[-0.008, 0.115]	0.80
G-Pre – G-Post (stage G)	0.061	0.023	0.051	[-0.004, 0.167]	0.80
Integration-stage effect (cell mean)	0.049	0.026	0.035	[-0.006, 0.108]	0.80
P-Pre – G-Pre (approach pre)	-0.028	-0.001	-0.036	[-0.139, 0.000]	0.60
Approach effect (cell mean)	-0.017	-0.005	-0.019	[-0.069, 0.002]	0.80
<i>Sensitivity: P-Post/G-Post with the fixed majority-vote rule ($\alpha = 0.50$)</i>					
P-Pre – P-Post (stage P)	0.180	0.025	0.018	[0.003, 0.828]	1.00
G-Pre – G-Post (stage G)	0.205	0.018	0.052	[0.007, 0.817]	1.00
Integration-stage effect (cell mean)	0.193	0.021	0.035	[0.005, 0.822]	1.00

development-ARI selection lands beyond the fragmentation transition: the selected NCVR-2P G-Pre graph (name-weighted, $s = 0.85$) retains only ≈ 195 edges on the development side, splitting into $\approx 2,205$ components with an LCC ratio of 0.0015 and component purity 0.99 — manifestly *not* one giant component — and analogously on test (LCC ratio 0.0019, purity 0.995). At that operating point G-Pre and P-Pre choose equivalent structures and produce identical partitions (ARI 0.952). The residual distance from a perfect partition is therefore not percolation of the selected graph; it is residual false-edge contamination — roughly 6–10% of retained edges are false (edge precision 0.90 development, 0.94 test) — combined with boundary sensitivity: one grid step below the selected point ($s = 0.75$) the equal-weight graph still carries a large, impure component (LCC ratio 0.115, purity 0.30), so accuracy changes sharply across a narrow sparsification band. The same transition appears benignly on the other datasets: FEBRL1, Restaurant, and DBLP-ACM cross from a percolated to a fragmented, high-purity graph between $s = 0.35$ and $s = 0.55$, and the graph cells then match the pairwise cells.

8.2 Why the majority-vote rule collapses post-clustering integration on NCVR-2P (RQ5, H2)

Under the fixed majority-vote rule the post-clustering cells fail on NCVR-2P for a reason visible without any test label — and the failure belongs to the interaction of coarse views with the *consensus rule*, not to view coarseness alone; it is neither structural nor inevitable. A per-view partition can never be finer than the equivalence classes of the view’s values: NCVR-2P has 1,978 distinct suburb strings across 4,000 records (the most frequent shared by 171 records) and 1,573 distinct postcodes (most frequent shared by 20). Suburb and postcode partitions are therefore necessarily far coarser than the 3,696 entities, and any pair of records sharing both a suburb cluster and a postcode cluster carries a 2-of-3 co-membership majority regardless of name. Crucially, at $\alpha = 0.5$ this majority forms an edge under *every* weight vector in the search space, because the smallest renormalized weight of any two views is 0.50: no admissible weighting lets the name view veto a suburb–postcode coalition. The consensus vote graph inherits these false edges — its test diagnostics at $\alpha = 0.5$ show component purity 0.80 but edge precision only 0.136 (Table 11) —

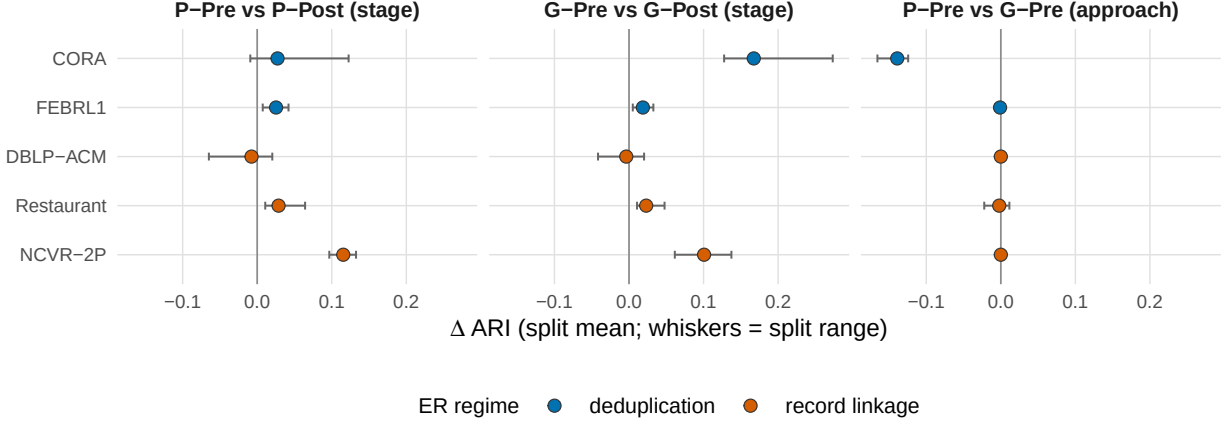


Figure 3: Paired held-out ARI differences by dataset under the tuned-method estimand (point = split mean, whiskers = split range, within-task dispersion). Panels show the two integration-stage contrasts (P-Pre–P-Post, G-Pre–G-Post) and the within-pre approach contrast (P-Pre–G-Pre); positive values favor the first-named cell.

and connected components over it merge entities, yielding $\text{ARI} \leq 0.136$. Pre-clustering pipelines avoid this because a pair needs jointly high *continuous* similarity across views before any clustering commitment is made: the same weight vectors that doom P-Post at $\alpha = 0.5$ let P-Pre reach 0.952. Consistent with this mechanism, development selection itself escapes the trap when allowed to: on this task it chooses $\alpha = 0.60$ with a name-weighted vector in every split and cell — a setting in which a suburb–postcode coalition no longer reaches the vote threshold — recovering P-Post and G-Post to ARI 0.837 and 0.851 (Section 7), still about 0.10 below the pre-clustering cells because the per-view name partition commits errors that consensus cannot undo. On datasets whose single views are individually discriminative (FEBRL1 addresses, Restaurant phones, DBLP-ACM titles), the per-view partitions are already fine, and the cost of clustering first is 0.003–0.025 ARI (occasionally mildly negative).

8.3 Complementarity descriptors and integration gain (RQ4, H5)

Table 12 reports the development-side descriptors alongside the held-out ARI gain, and Table 13 their rank associations with the gain. Because a dataset is the independent unit, descriptors are aggregated to dataset means and the associations are computed over $n = 5$ points; with five observations a Spearman coefficient is an exploratory, descriptive quantity, and we present it as nothing more. Three associations stand out: gain is higher where all views are jointly observed (pairs-with-all-views rate, $\rho = 0.90$; minimum view availability, $\rho = 0.89$), lower where the best single view is already strong ($\rho = -0.70$), and lower where view similarities are mutually correlated ($\rho = -0.70$). These match the CORA case: its venue view is missing for roughly one in seven labeled records, a quarter of record pairs lack at least one view, and its title view alone nearly solves the task, so integration gains little that title does not already provide. We present these as observable, outcome-independent signals that practitioners can compute before committing to a multiview pipeline — not as a predictive model.

Table 8: Per-dataset paired held-out ARI differences under the tuned-method estimand (split mean, SD, and range over the five entity-disjoint splits); supplementary detail behind Table 7.

Dataset	Contrast	Mean Δ ARI	SD	Split range
CORA	P-Pre – P-Post	0.027	0.054	[-0.009, 0.123]
	G-Pre – G-Post	0.167	0.061	[0.128, 0.273]
	P-Pre – G-Pre	-0.139	0.018	[-0.166, -0.124]
FEBRL1	P-Pre – P-Post	0.025	0.016	[0.007, 0.042]
	G-Pre – G-Post	0.019	0.010	[0.005, 0.033]
	P-Pre – G-Pre	-0.001	0.004	[-0.008, 0.002]
DBLP-ACM	P-Pre – P-Post	-0.008	0.034	[-0.065, 0.020]
	G-Pre – G-Post	-0.004	0.024	[-0.042, 0.020]
	P-Pre – G-Pre	0.000	0.000	[0.000, 0.000]
Restaurant	P-Pre – P-Post	0.029	0.022	[0.011, 0.064]
	G-Pre – G-Post	0.023	0.017	[0.011, 0.048]
	P-Pre – G-Pre	-0.002	0.012	[-0.022, 0.012]
NCVR-2P	P-Pre – P-Post	0.115	0.016	[0.097, 0.133]
	G-Pre – G-Post	0.101	0.034	[0.061, 0.137]
	P-Pre – G-Pre	0.000	0.000	[0.000, 0.000]

Table 9: Held-out ARI gain of the development-selected best multiview cell over the development-selected best single view (split mean, SD, and range).

Dataset	Best SV ARI	Gain	SD	Split range	Best cell	SV view
CORA	0.898	-0.054	0.128	[-0.271, 0.060]	G-Pre	title
FEBRL1	0.905	0.089	0.010	[0.080, 0.106]	P-Pre	address
DBLP-ACM	0.917	-0.004	0.019	[-0.037, 0.010]	P-Pre	title
Restaurant	0.964	0.028	0.014	[0.011, 0.048]	P-Pre	phone
NCVR-2P	0.569	0.383	0.047	[0.303, 0.424]	P-Pre	name

9 Robustness and Sensitivity

Selection optimism. Development-to-test ARI gaps of the selected configurations (shared grid; fixed rule for the post-clustering cells, which coincides with their modal development-selected α on CORA) are small on four datasets (absolute split-mean gaps below 0.05, often negative because the smaller test sides present easier pair imbalance). CORA is the exception: all four cells are optimistic, with mean gaps of +0.227 (P-Pre), +0.254 (P-Post), +0.254 (G-Post), and +0.091 (G-Pre). With only 118 entities, a 60/40 entity split leaves heterogeneous sides, and sharply tuned thresholds transfer poorly — while G-Pre’s community structure transfers best. This is direct, within-study evidence that unmatched or leaky selection protocols could have reversed cell rankings: on CORA the development ranking (all four cells at ARI 0.986–0.990) bears little resemblance to the test ranking (G-Pre first by 0.14).

Sensitivity to hyperparameters. The response surfaces are sharp. The fraction of a cell’s 98 configurations within 95% of its best development ARI ranges from 0.6% (G-Post on NCVR-2P) to 38% (G-Pre on FEBRL1); on the linkage datasets the median configuration of the post-clustering cells is near zero. Threshold instability is thus a real failure mode (RQ5): accuracy depends on landing in a narrow region of the grid, which is precisely why matched base-configuration grids are

Table 10: Mean runtime per configuration (seconds) and failed-configuration counts.

Dataset	P-Pre	P-Post	G-Pre	G-Post	Failed
CORA	0.03	0.12	0.08	0.09	0
FEBRL1	0.00	0.02	0.01	0.02	0
DBLP-ACM	0.04	0.32	0.07	0.20	0
Restaurant	0.00	0.01	0.01	0.01	0
NCVR-2P	0.21	0.76	0.82	0.52	0

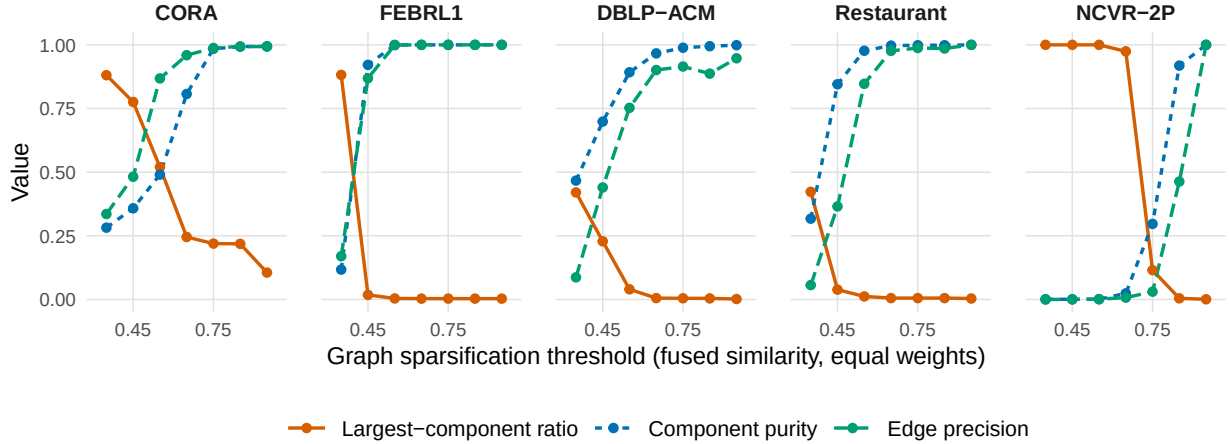


Figure 4: Percolation diagnostic on development sides (equal weights, resolution 1.0; mean over splits; sparsification 0.35–0.95). Where the largest-component ratio is high while purity and edge precision are low, the retained graph has merged entities through false edges. Selected configurations sit beyond each dataset’s fragmentation transition.

necessary for a fair comparison.

Selection-metric agreement. Selecting configurations by the supplementary development pairwise F_1 instead of development ARI would have chosen the identical configuration in all 100 (dataset, split, cell) selections. On these tasks the two metrics rank the search space’s optima identically, so the primary conclusions do not hinge on the choice between them; ARI remains the appropriate primary metric because the system’s output is a partition.

Grid-boundary occupancy. Neither pre-clustering cell ever selected a grid edge: all 50 P-Pre and G-Pre selections are interior (NCVR-2P selects $t = 0.85$ and $s = 0.85$ with the grids extending to 0.95), so the reported P-Pre and G-Pre ceilings are not grid-capped. The post-clustering cells selected the top of the threshold or sparsification grid in 18 of 50 selections (both cells on Restaurant in four splits and on DBLP-ACM in two; G-Post on NCVR-2P in all five and P-Post in one). For the post-clustering cells this reflects a monotone preference for the finest available per-view partitions — a structural property of clustering views first — rather than a truncated optimum for a smooth response surface.

Consensus-threshold sensitivity. Table 14 and Figure 6 unfold the α dimension that the primary selection searches: for each $\alpha \in \{0.50, 0.60, 0.67, 0.75\}$ the full 98-configuration grid is searched on development entities and the development-ARI-selected configuration is evaluated once on test; α is never chosen on test results, and the primary P-Post/G-Post rows in Table 4 correspond to the development-selected column of this sweep. On four datasets the effect of α is modest

Table 11: Graph diagnostics of the selected configurations on test sides (mean over splits). Density is retained edges over the side’s candidate-pair count ($m(m-1)/2$ for deduplication, $n_A \times n_B$ for linkage). For G-Post the diagnosed graph is the consensus vote graph that determines the final partition.

Dataset	Cell	Retained edges	Density	LCC ratio	Comp. purity	Edge prec.	Edge recall
CORA	G-Pre	18,687	0.071	0.228	0.837	0.924	0.872
	G-Post	17,980	0.068	0.241	0.828	0.924	0.837
FEBRL1	G-Pre	200	0.003	0.007	0.998	0.998	0.998
	G-Post	199	0.002	0.009	0.995	0.981	0.977
DBLP-ACM	G-Pre	893	0.001	0.005	0.987	0.949	0.953
	G-Post	904	0.001	0.005	0.981	0.934	0.951
Restaurant	G-Pre	44	0.002	0.006	0.999	0.996	0.995
	G-Post	45	0.002	0.007	0.997	0.974	0.986
NCVR-2P	G-Pre	127	2.0e-04	0.002	0.995	0.942	0.979
	G-Post	482	0.001	0.006	0.796	0.136	0.536

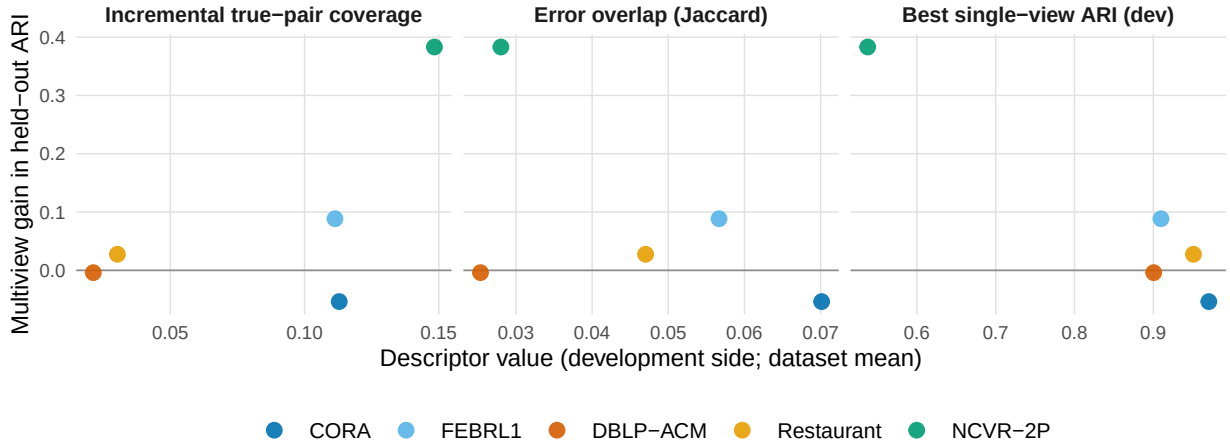


Figure 5: Held-out multiview ARI gain versus three development-side complementarity descriptors (points = dataset means; $n = 5$; exploratory).

(held-out ARI changes by at most 0.05 across the grid, with the test-best value task-dependent), and development selection lands on or near the test-best value on CORA, DBLP-ACM, FEBRL1, and NCVR-2P; on Restaurant one split’s development choice of $\alpha = 0.60$ (P-Post) costs 0.064 ARI against the fixed rule, a visible but bounded selection-noise price. On NCVR-2P the effect is decisive: moving from $\alpha = 0.50$ to 0.60 lifts P-Post from 0.124 to 0.837 and G-Post from 0.136 to 0.851, because selection switches to a name-weighted vector under which a suburb-postcode coalition no longer reaches the vote threshold (Section 8), and development selection finds exactly this setting in every split. The fixed-rule collapse is therefore a property of the majority-vote constant, not of the integration stage per se; yet even at the best α , P-Post and G-Post remain about 0.10 ARI below the pre-clustering cells on this task.

10 Discussion

Answering the opening question. The integration stage and consensus calibration interact, and the answer must be stated in those terms. Pre-clustering integration is robust without a

Table 12: Development-side complementarity descriptors (dataset means) and held-out multiview ARI gain.

Dataset	View corr.	Error overlap	Incr. coverage	Best SV dev ARI	ARI gain
CORA	0.496	0.070	0.113	0.970	-0.054
FEBRL1	0.086	0.057	0.111	0.910	0.089
DBLP-ACM	0.042	0.025	0.021	0.901	-0.004
Restaurant	0.240	0.047	0.030	0.951	0.028
NCVR-2P	0.036	0.028	0.148	0.538	0.383

Table 13: Spearman rank association between development-side descriptors (dataset means) and held-out multiview ARI gain ($n = 5$ datasets; exploratory, not predictive).

Descriptor (computed on development data)	Spearman ρ ($n = 5$)
Mean between-view similarity correlation	-0.70
Mean per-pair view disagreement	0.20
Minimum view availability	0.89
Pairs with all views observed	0.90
Best single-view ARI (dev)	-0.70
Weakest single-view ARI (dev)	-0.30
Mean error overlap (Jaccard)	-0.30
Union true-pair coverage	-0.10
Incremental coverage over best view	0.40

consensus parameter: it matched or outperformed calibrated post-clustering integration on four of five datasets (median cell-mean advantage 0.026 ARI, the median over datasets of $\overline{\text{Pre}} - \overline{\text{Post}}$; the per-approach paired medians are 0.027 and 0.023) and never trailed it by more than 0.008. Post-clustering integration can be competitive — essentially tying or marginally leading where every view is individually strong (DBLP-ACM) — but it is sensitive to the co-membership threshold: the same pipelines span ARI 0.124–0.837 (P-Post) on one task depending only on α . When a task combines structurally coarse views with the fixed majority-vote rule, the post-clustering cells fail catastrophically and observably; both ingredients of that failure are diagnosable before any experiment (view coarseness from value multiplicity; the rule’s inability to veto coarse-view coalitions from the weight grid and α alone), and development-side calibration of α repairs most of the collapse while leaving a residual deficit near 0.10. The practical default that follows is pre-clustering integration when views can be integrated — not because clustering views first must fail, but because it removes a calibration burden that can fail badly at its default setting; post-clustering integration remains attractive when views cannot be integrated (for example, views held by different parties), provided α is treated as a tuned parameter.

Hypotheses revisited. H1 (a systematic main effect of the base clustering approach) is *not supported*, in an instructive way: the two implemented methods were nearly indistinguishable on four datasets (cell-mean ARI differences ≤ 0.007) and neither dominated — within post-clustering integration the threshold method was marginally ahead on CORA (+0.001) and DBLP-ACM (+0.004), differences far too small to be material — and the approach effect is conditional on the integration stage (approach $\times$ stage interaction carried by CORA): it appears only within pre-clustering integration and only there, where the separation (0.139) is a selection-transfer effect rather than a difference in attainable accuracy. There is no dataset here on which thresholded connectivity materially beats the Louvain method. H2 must be revised rather than simply accepted:

Table 14: Consensus-threshold robustness: held-out ARI of the development-selected P-Post and G-Post configurations as a function of the consensus vote threshold α (mean over five splits; α never selected on test results).

Dataset	Cell	$\alpha = 0.50$	$\alpha = 0.60$	$\alpha = 0.67$	$\alpha = 0.75$
CORA	P-Post	0.732	0.757	0.777	0.777
	G-Post	0.732	0.757	0.776	0.776
FEBRL1	P-Post	0.969	0.964	0.953	0.953
	G-Post	0.977	0.980	0.969	0.969
DBLP-ACM	P-Post	0.912	0.914	0.923	0.923
	G-Post	0.909	0.914	0.918	0.918
Restaurant	P-Post	0.975	0.935	0.926	0.926
	G-Post	0.975	0.935	0.935	0.935
NCVR-2P	P-Post	0.124	0.837	0.837	0.837
	G-Post	0.136	0.851	0.851	0.851

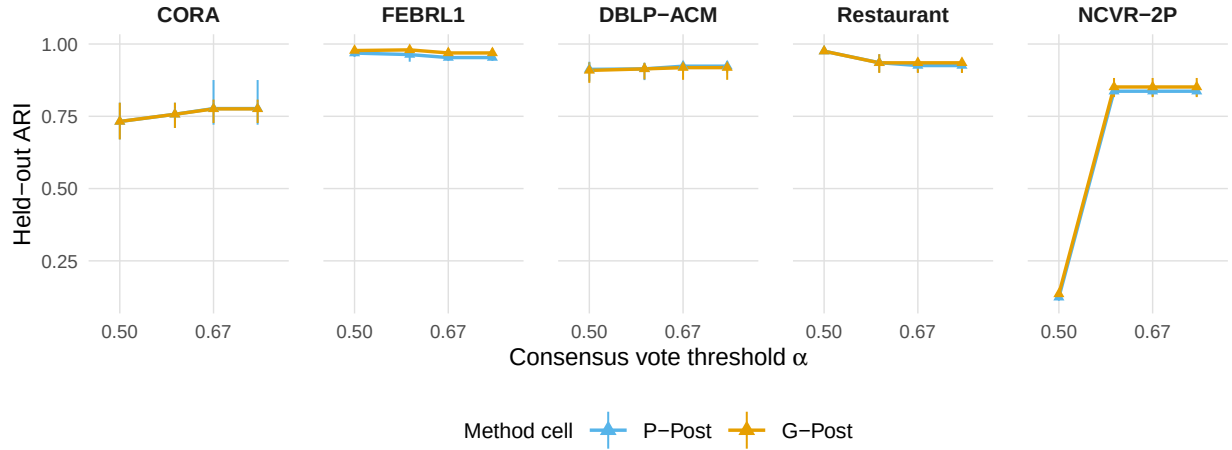


Figure 6: Held-out ARI of the post-clustering cells (P-Post, G-Post) as the consensus vote threshold α varies (point = split mean, whiskers = split range).

under the tuned-method estimand the integration-stage effect is positive on four of five datasets (median 0.026 ARI) but not unanimous, and the losses of clustering views first are not primarily a story of views being “individually weak but complementary.” What the data show is an interaction between the integration stage and consensus calibration: per-view partitions do commit errors that consensus cannot undo (the residual 0.10 deficit on NCVR-2P at the best development-selected α), but the catastrophic losses arise only when structurally coarse views meet the fixed majority-vote rule, and calibrating α on development data repairs most of them. Our data cannot separate “complementarity” from “individual view weakness” as the driver of the residual deficit, since NCVR-2P exhibits both. H3 is only partially assessable: regime-level differences exist descriptively, but the larger linkage-side integration-stage effect is carried entirely by NCVR-2P (without it the two remaining linkage tasks show smaller stage effects than the deduplication tasks), so no stable three-way interaction is claimed. H4 was not directly tested at the selected operating points, because the development-selected Louvain configurations did not materially degrade and operated beyond the percolation region. The proposed false-edge percolation mechanism was nevertheless observed along the unselected low-sparsification path, where large impure components formed

(high largest-component ratio with low component purity and edge precision) before the graph fragmented at higher sparsification thresholds; the residual accuracy loss of the selected graph configurations comes instead from remaining false edges and the sharpness of the transition, not from operating on a giant component. We therefore distinguish the performance of the selected configurations from the mechanism observed along unselected parameter paths, and do not use the latter as direct evidence for the former. The observations are descriptively consistent with H5, but $n = 5$ ($|\rho| = 0.70\text{--}0.90$) is insufficient for confirmatory support or prediction; H5 receives preliminary descriptive support and remains exploratory.

Neither approach dominates. A noteworthy result is that the choice of base clustering approach mattered far less than either the integration stage or the selection protocol. The two implemented methods were nearly indistinguishable on four datasets (cell-mean ARI differences ≤ 0.007): the pre-clustering cells produce identical partitions on DBLP-ACM and NCVR-2P, and neither method dominated — within post-clustering integration the threshold method was marginally ahead on CORA and DBLP-ACM and Louvain marginally ahead elsewhere, all immaterial. Their only material separation was on CORA under pre-clustering integration, where community structure transferred across the entity split (gap 0.09) while sharply tuned thresholds did not (gap 0.23), giving Louvain a 0.139 ARI lead. Practitioners tuning on limited labeled data may prefer the Louvain graph cells for this flatter generalization gap; we found no setting in which thresholded connectivity was materially the better choice, though this comparison covers only Louvain and not the wider graph-clustering family.

When is multiview integration worth it? The answer our data support is conditional and computable in advance, though from only five tasks: integration pays when no single view is strong ($\rho = -0.70$ between best-single-view strength and ARI gain), when views are jointly observed ($\rho = 0.89\text{--}0.90$), and when views do not duplicate each other’s similarity structure ($\rho = -0.70$). On two of five datasets the honest answer is that integration bought nothing that the best single view did not already provide (gains -0.004 and -0.054); reporting this is part of the study’s purpose.

Selection protocol as a first-class result. The CORA case demonstrates within one controlled study how a comparison performed without held-out entities would have concluded that all four cells are indistinguishable (development ARI 0.986–0.990 each), where the held-out ranking actually separates G-Pre from P-Pre by 0.14. Matched-grid, entity-disjoint selection is not methodological pedantry; on small benchmarks it changes the ranking. The consensus-threshold analysis makes the same point for rule constants: a single fixed α can flip a method cell from unusable to competitive, which is why the primary protocol gives consensus constants the same searched-on-development, held-out-tested treatment as thresholds and weights.

11 Limitations

Datasets. Five benchmark tasks — two deduplication, three linkage, three of them bibliographic or commercial directory data and two derived from synthetic corruption — cannot represent the full diversity of ER problems. The dataset is the independent unit here ($n = 5$); repeated entity-disjoint splits are dependent, overlapping resamples of one record universe and quantify within-task uncertainty only, which is why split dispersion is reported as mean, SD, and range rather than as confidence intervals. The large-magnitude findings rest on single datasets (NCVR-2P for the consensus-rule collapse and the residual order deficit, CORA for selection optimism) and should be generalized only as mechanisms, not as universal effect sizes; headline averages are therefore accompanied by medians and leave-NCVR-out values throughout.

Truth quality and provenance. CORA’s evaluation universe is restricted to the 1,815

records present in its transitively closed truth graph; the 64 excluded records cannot be confirmed as singletons. FEBRL1 and NCVR-2P inherit the corruption model of their generators, which may understate real-world error correlation across views. NCVR-2P is a fixed, prepared NCVR-derived snapshot distributed with the study (checksum-pinned); its exact subset-construction procedure — party selection, record and matched-entity sampling, and any random seed — is not recorded and could not be reconstructed, and the file is not a clean subset of any bundled multi-party NCVR source. Results on it are therefore reproducible from the released file but the task itself is not independently reconstructible from public sources, and we treat it as a fixed prepared task rather than a standard reproducible benchmark.

View definitions. Results are conditional on the three-view definitions and the fixed per-view similarity functions declared in the protocol. Different view partitions of the same schemas (or learned similarities) could shift both the order effect and the integration gain.

Method coverage. The graph family is represented by Louvain with two resolutions over thresholded similarity graphs, and consensus by a weighted co-membership vote with a searched-in-robustness threshold. Leiden, correlation clustering, k -nearest-neighbor sparsification, and richer consensus schemes are plausible alternatives that this study does not cover; conclusions about “graph/community clustering” are conclusions about this Louvain representative, not the family’s frontier. The pre-clustering cells never selected a grid edge, but the post-clustering cells sit at the top of their grids in 18 of 50 selections (a structural preference for the finest per-view partitions), so post-clustering ceilings remain conservative estimates.

Search-space asymmetry. Equal candidate counts on the shared grid do not make the four cells fully symmetric: the post-clustering cells carry an additional consensus parameter, and selecting it on development data gives them a four-fold larger effective search space in the tuned-method comparison. We keep the two estimands separate and label every quantity, but readers should interpret the tuned-method comparison as “a calibrated post-clustering pipeline against a calibration-free pre-clustering pipeline,” not as a comparison of equally-sized search spaces.

Model-selection uncertainty. With 98 shared-grid configurations per cell (times four consensus thresholds for the post-clustering cells) and five splits, selection noise remains visible (notably on CORA, and in one Restaurant split’s consensus-threshold choice); larger budgets or nested selection could tighten these estimates but were outside the computational envelope of this study.

Scale. Exhaustive candidate generation (pair completeness 1.0) was feasible at these sizes and deliberately removes blocking as a confounder; conclusions do not automatically transfer to corpus sizes where blocking is unavoidable and candidate-generation loss interacts with the design dimensions.

12 Conclusion

We presented a controlled, selection-aware experimental study of multiview entity resolution that crosses the view-specific base clustering approach with the view-integration stage relative to clustering under both ER regimes, with held-out ARI as the primary outcome and the consensus threshold of the post-clustering pipelines treated as a tuned parameter. The two axes interact, so their effects are reported conditionally. The integration stage and consensus calibration interact: pre-clustering integration is robust without any consensus parameter and matched or outperformed calibrated post-clustering integration on four of five datasets (median cell-mean advantage 0.026 ARI), while post-clustering integration can be competitive — occasionally marginally ahead — but is sensitive to the co-membership threshold, whose fixed majority-vote default fails catastrophically

exactly where views are structurally coarse. That failure is diagnosable in advance (value multiplicity; no admissible weighting can veto a coarse-view coalition at $\alpha = 0.5$) and is largely, though not fully, repaired by development-side calibration of α . The base clustering approach has no material main effect, and what effect exists is conditional on the integration stage: the two implemented methods were nearly indistinguishable except on CORA (the benchmark with the fewest truth entities) under pre-clustering integration, where community structure generalized across the selection split better than sharply tuned thresholds (a 0.14 ARI lead for Louvain), with post-clustering differences elsewhere small and immaterial; false-edge percolation is real and observable in graph diagnostics along the lower-sparsification path, while selected graph configurations operate beyond the fragmentation transition and their residual losses come from remaining false edges and boundary sensitivity. Multiview integration is not automatically worthwhile: its held-out ARI gain over the best single view spans -0.054 to $+0.383$ (cross-dataset median 0.028) and is associated with development-side descriptors that practitioners can compute before committing. All analyses, results, tables, and figures are regenerable from the released scripts, frozen protocol, and checksum-pinned input files; the prepared NCVR-2P input is reproducible as a released fixed file but is not independently reconstructible from the available public NCVR sources.

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